MES COLLEGE OF ARTS, COMMERCE AND SCIENCE

DEPARTMENT OF COMPUTER APPLICATIONS

PROJECT SYNOPSIS

ON

INVORA

AI-Powered Personal Learning Assistant

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PROJECT OVERVIEW

Artificial Intelligence (AI) is transforming education by making learning more interactive and efficient. However, students often spend a lot of time reading lengthy PDF study materials, preparing notes, and creating practice questions manually.

INVORA (Intelligent Virtual Online Resource Assistant) is an AI-powered personal learning assistant designed to simplify and enhance the learning process. The application enables students to upload PDF study materials and interact with them using Artificial Intelligence. By leveraging the Google Gemini API, INVORA provides instant answers to questions based on uploaded documents, automatically generates concise notes and summaries, creates quizzes for self-assessment, and monitors learning progress through an interactive dashboard.

The primary goal of INVORA is to reduce students' manual effort, improve learning efficiency, and provide a smart, user-friendly platform for academic success. The application combines modern web technologies with AI to deliver an interactive and personalized learning experience suitable for students, educational institutions, and self-learners.

Hardware and Software Requirements

Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 or above
RAM	8 GB
Storage	256 GB SSD/HDD
Internet	Required

Software Requirements

Software	Technology
Operating System	Windows 10/11
Frontend	HTML, CSS, JavaScript
Backend	Python (Flask)
Database	MySQL
AI Integration	Google Gemini API
IDE	Visual Studio Code

Problem Statement

Students widely use PDF notes and digital study materials for learning, but finding relevant information, preparing notes, and creating practice questions manually can be time-consuming and inefficient. Existing learning platforms provide study resources but do not offer AI-powered interaction with uploaded documents or personalized learning support.

INVORA (Intelligent Virtual Online Resource Assistant) addresses these challenges by allowing students to securely upload and manage PDF study materials, interact with them using AI, generate notes and summaries, create quizzes for self-assessment, and track their learning progress through an interactive dashboard. The system aims to simplify learning, save time, and improve academic performance.

Existing System

At present, students use PDF notes, e-books, and online study materials for learning. They manually search through lengthy documents to find relevant information, prepare notes by themselves, and rely on different websites or applications to create quizzes and assess their knowledge. These methods require considerable time and effort, making the learning process less efficient.

Limitations of Existing System

- Students spend more time searching for information in PDF documents.
- Notes and summaries are prepared manually.
- No AI-based interaction with uploaded study materials.
- Students need separate platforms for quizzes and revision.
- No centralized system to monitor learning progress.

Proposed System

The proposed system is an AI-powered personal learning assistant designed to improve the learning experience of students. It provides a centralized platform where users can securely upload and manage PDF study materials, interact with the content using Artificial Intelligence, generate notes and summaries, create quizzes for self-assessment, and monitor their learning progress. By integrating the Google Gemini API with modern web technologies, the application simplifies studying, reduces manual effort, and supports effective exam preparation.

The system combines multiple learning tools into a single platform, eliminating the need to use different applications for note-taking, quiz preparation, and progress tracking.

Objectives

- To develop an AI-powered personal learning assistant.
- To provide secure user registration and login.
- To enable students to upload and manage PDF study materials.
- To generate AI-based notes and summaries.
- To create quizzes for self-assessment.
- To track students' learning progress through a dashboard.

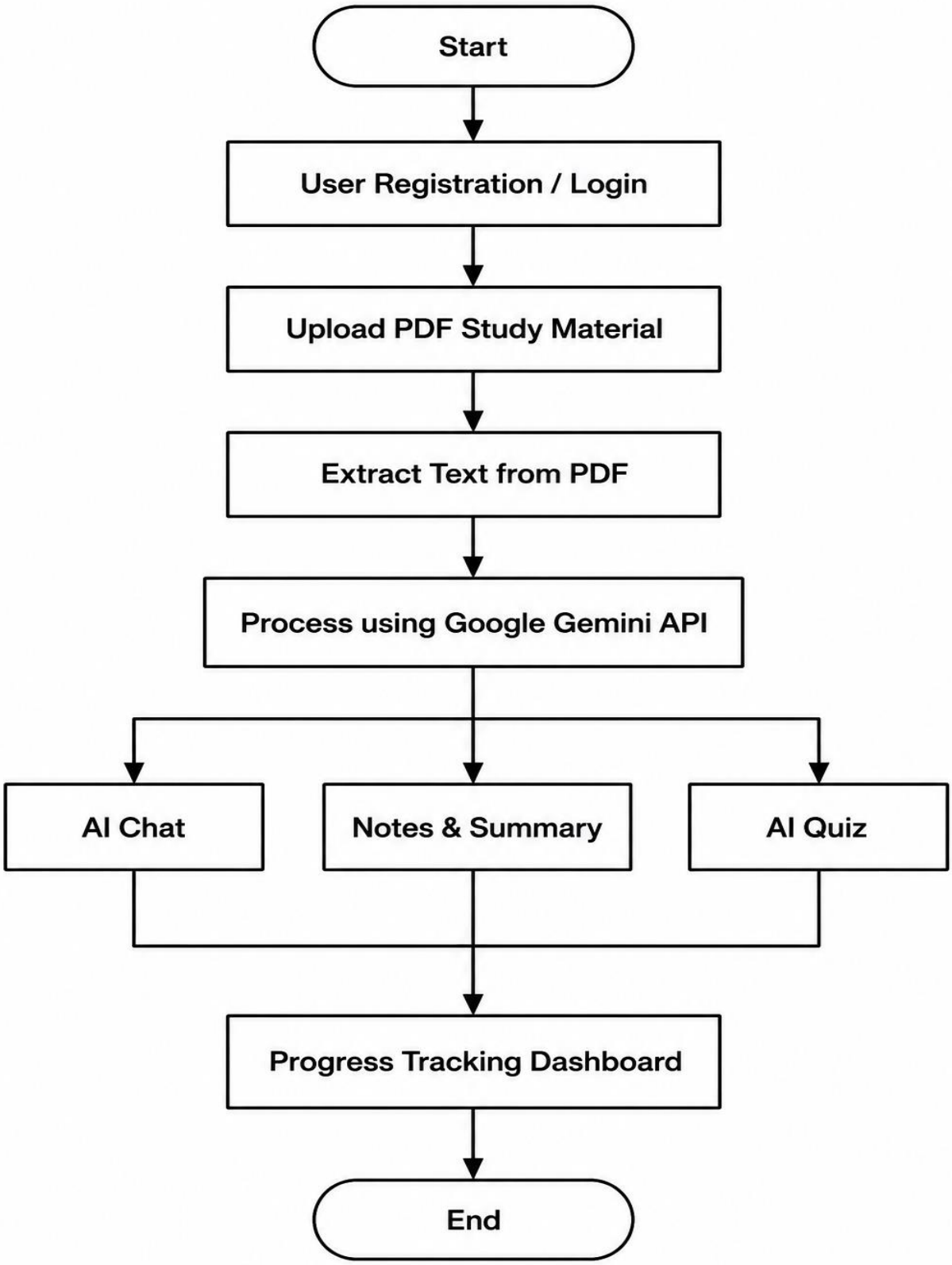
Key Features

1. User Login & Registration
2. PDF Upload & Management
3. AI Chat with Uploaded PDFs
4. AI Notes & Summary Generation
5. AI Quiz Generation
6. Progress Tracking Dashboard

Methodology

The proposed system follows a systematic approach to provide an AI-powered learning platform for students. Initially, users register and log in to the application to access their personal workspace. They can then upload PDF study materials, which are processed to extract the text content. The extracted information is analyzed using the Google Gemini API to enable intelligent interactions.

The system allows users to ask questions related to the uploaded PDFs, automatically generates notes and summaries, and creates quizzes for self-assessment. Finally, the application records user activities and quiz performance in a progress tracking dashboard, helping students monitor their learning progress effectively.


Start
User Registration / Login
Upload PDF Study Material
Extract Text from PDF
Process using Google Gemini API
AI Chat
Notes & Summary
AI Quiz
Progress Tracking Dashboard
End